



SWITCHES

SYNOPSIS

INTRODUCTION

DESCRIPTION

WORKING PRINCIPLE

DIFFERENT TYPES

1.ELECTROMECHANICAL P.S

2.SOLID-STATE P.S

OTHER TYPES

SELECTION OF PRESSURE SWITCH

APPLICATIONS

- ❖ **Switches** are essentially binary devices that are either completely on ("closed") or completely off ("open").
- ❖ There are many ***different types of switches***, like
 - ❖ Pressure switches,
 - ❖ Electrical switches,
 - ❖ Temperature switches
 - ❖ Speed switches etc.

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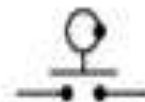
1.ELECTROMECHANICAL P.S

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OTHER TYPES

SELECTION OF PRESSURE SWITCH

APPLICATIONS

Toggle switch 	Pushbutton switch 
Selector switch 	Joystick switch 
Lever actuator limit switch 	Proximity switch 
Speed switch 	Pressure switch 

PRESSURE SWITCHES

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- ❖ Pressure switches are devices that are configured to sense a change in pressure and respond in a specified manner.
- ❖ The pressure switch is utilized in many different environments, including
 - ❖ Manufacturing machinery
 - ❖ Utility plants
 - ❖ Public buildings

Description

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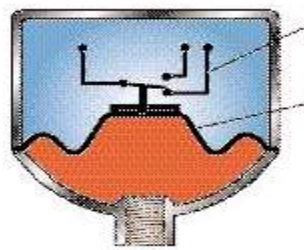
2.SOLID-STATE P.S

OTHER TYPES

SELECTION OF PRESSURE SWITCH

APPLICATIONS

- ❖ **Contacts** :The conductive parts in a switch used to make and break the electrical connection are called contacts.
- ❖ **Set point of switch** :The pressure at which it turns ON or OFF is called Set point of switch.
- ❖ **Pressure range** :The range of pressure that is needed to be switched at.
- ❖ **Circuit completion** :The pressure is fed inside the bellows which causes the contact of the plate with the circuit and current flows which either on the alarm or send signal to control center.
- ❖ **Primary sensor** : A Bourdon tube, diaphragm or bellows can be used to actuate the switch.



Force

Contacts

Diaphragm

ON

OFF

Working Principle

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❖ **Open switch** :The contact in a normally open switch remains opened until the pressure rises above the set point. Then the sensing element makes the contacts snap to close position. The contacts open again when the pressure falls below the set point.

❖ **Closed switch**: The contacts in a normally closed switch remain closed until the pressure rises above the set point. Then the contacts snap open and remain open until a pressure drops below the set point again.

TYPES OF PRESSURE SWITCHES

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BASIC TYPES:

- ❖ Electromechanical pressure switch
- ❖ Solid state pressure switch

Electromechanical Pressure Switches

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The most common electromechanical pressure switches are composed of a

- ❖ Sensing element
- ❖ Electrical snap-action switch.

Solid-State Pressure Switches

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Solid-state pressure switches contains following characteristics:

- ❖ One to four or more switch points,
- ❖ Digital displays,
- ❖ Analog and digital outputs,
- ❖ Full programmability.
- ❖ Provide a proportional 4-20 mA analog signal or digital output.



Other Types

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1. High Pressure Switch
2. Diaphragm Pressure Switch
3. Low Pressure Switch
4. Liquid Level Pressure Switch
5. Air Pressure Switch
6. Air Compressor Pressure switch

1. High Pressure Switch

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APPLICATIONS

This pressure switch is designed for the demanding requirements of ***lager, heavy duty commercial air compressors*** (up to 400 psi).

- ❖ Resists the harsh conditions
- ❖ Made up of stainless steel



2. Diaphragm Pressure Switch

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APPLICATIONS

The diaphragm pressure switch is available for various sumps, effluent, and sewage application to control the liquid level.

- ❖ For to turn-on or turn-off the pump
- ❖ More reliable than traditional float ball mechanism



3. Low Pressure Switch

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APPLICATIONS

- ❖ Widely used in food waste disposer, pumps for swimming pools, hot tubs, sanitary equipment, medical equipment etc.
- ❖ The low pressure switch has Operating Pressure Range ***20 mbar to 1000 mbar.***



4. Liquid Level Pressure Switch

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APPLICATIONS

The liquid level pressure switch can be used to control the level of water or other non-hazardous liquids.

- ❖ Basic functioning operation is similar to diaphragm pressure switch.
- ❖ Operable within the range of -10°C to 85°C .



6. Air Compressor Pressure switch

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- ❖ The air compressor pressure switch is used to regulate the tank pressure between two preset values on electrically driven air compressors.
- ❖ It has operating ***Pressure Range: 25-175 psi***



SELECTION OF PRESSURE SWITCH:-

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The following is a list of questions that need to be answered when we are going to select a specific type of pressure switch.

1.How Often Will The Switch Be Activated

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Electromechanical switches are subject to fatigue. But a solid-state switch is not subject to fatigue

Pressure Range

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Selecting the proper relationship between the switch point and the operating pressure range of a switch affects accuracy and life.

4. High-Pressure Spikes And Surges

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- ❖ Pressure surges and transient pressure spikes can greatly exceed the normal operating pressure of a system.
- ❖ It is not unusual for a switch to fail because the pressure spike exceeds the proof pressure

5. How Many Switch Points Are Needed

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- ❖ When sensing pressure at one point, only one switch point usually is required.
- ❖ However some systems require two or even four switch points to be monitored, controlled, or alarmed.

6. Set Point Adjustment

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APPLICATIONS

- ❖ In some applications, the set point is permanently fixed, while in others some adjustment is required.

APPLICATIONS

Pressure switches in their simple form detect a pressure change, and at a predetermined level or set point, open or close a contact. They act as.

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SELECTION OF PRESSURE SWITCH

APPLICATIONS

1. Safety devices
2. Alarms
3. Control elements within a system
4. Steam power plant
5. Electric motors
6. Gas compressor
7. Automobiles
8. Pumps
9. Boilers
10. Steam turbines

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